

Project Report

Team B:

Members:

1. Alfi
2. Liza
3. Shuchi

STROKE PREDICTION USING MACHINE LEARNING

1. Introduction:

Stroke is one of the leading causes of death and long-term disability worldwide. Early prediction of stroke risk can help medical professionals take preventive measures and improve patient outcomes. With the growth of healthcare datasets and machine learning techniques, it has become possible to analyze medical data and identify patterns associated with stroke occurrence. The purpose of this project is to analyze a healthcare stroke dataset and build a machine learning model capable of predicting whether a person is likely to experience a stroke. The dataset used contains several health-related attributes such as age, gender, hypertension, heart disease, BMI, glucose level, smoking status, and other demographic information.

2. Dataset Description:

The dataset contains medical and demographic information about patients. Each record corresponds to an individual patient.

Important attributes:

Feature	Description
age	Age of the patient
gender	Gender of the patient
hypertension	Whether the patient has high blood pressure
heart_disease	Whether the patient has heart disease
ever_married	Marital status
work_type	Type of employment
Residence_type	Urban or rural residence
avg_glucose_level	Average blood glucose level
bmi	Body Mass Index
smoking_status	Smoking behavior
stroke	Target variable (1 = stroke, 0 = no stroke)

3. Data Preprocessing:

Raw healthcare data often contains missing values, irrelevant columns, or inconsistent formats. Therefore, preprocessing is necessary before applying machine learning algorithms.

Removing Unnecessary Columns

The **ID column** does not contribute to prediction and is removed.

```
df.drop('id', axis=1, inplace=True)
```

Handling Missing Values

The dataset contains missing values in the **BMI column**. These values are replaced with the median value.

```
df['bmi'].fillna(df['bmi'].median(), inplace=True)
```

Encoding Categorical Data

Machine learning models require numerical input. Therefore categorical features such as gender, smoking status, and work type are converted using **Label Encoding**.

```
from sklearn.preprocessing import LabelEncoder
```

```
lb = LabelEncoder()
```

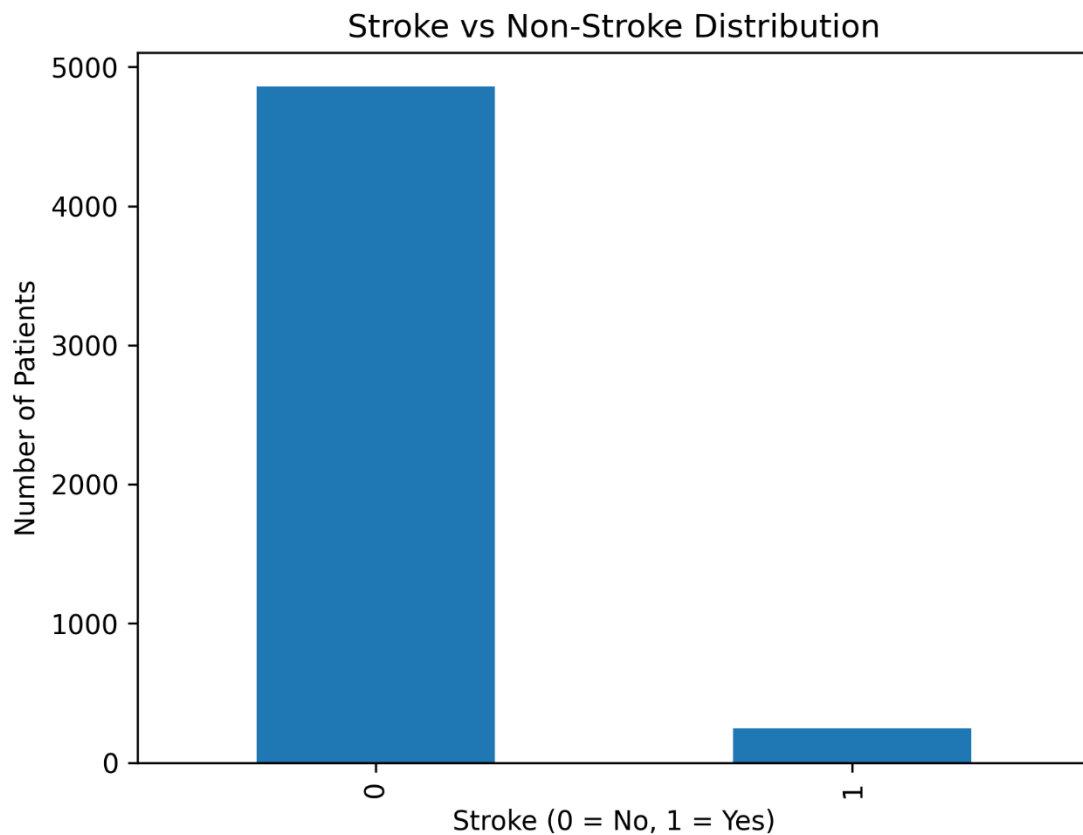
```
df = df.apply(lb.fit_transform)
```

4. Data Visualisation:

Data visualization helps to understand the relationships between variables and detect patterns within the dataset.

Stroke Distribution:

A bar chart shows how many individuals experienced a stroke.



Age vs Stroke:

Older individuals have a higher likelihood of stroke.

5. Feature Engineering:

Feature engineering helps create useful insights and prepare the dataset for machine learning. A correlation heatmap helps identify relationships between variables.

6. Feature Selection:

Feature selection is used to determine the most important features for prediction.

Selected Features

The most relevant features include:

- Age
- Work type
- Average glucose level
- BMI
- Smoking status

These features are used to train the machine learning model.

7. Handling Class Imbalance:

Since the dataset contains far more non-stroke cases, the **SMOTE** (Synthetic Minority Over-sampling Technique) method is used to balance the dataset.

8. Machine Learning Model Generation:

The dataset is split into training and testing sets. The Random Forest Classifier is used to build the prediction model.

9. Model Evaluation:

Accuracy:

The Random Forest model achieved high prediction accuracy.

```
from sklearn.metrics import accuracy_score
accuracy = accuracy_score(y_test, y_predict)
print("Accuracy:", accuracy)
```

Classification Report:

This report includes:

1. Precision
2. Recall
3. F1 Score

```
from sklearn.metrics import classification_report
```

```
print(classification_report(y_test, y_predict))
```

Confusion Matrix:

The confusion matrix shows how well the model correctly predicts stroke and non-stroke cases.

10.Conclusion:

This project demonstrates how machine learning can be applied to healthcare datasets to predict stroke risk. The workflow included data preprocessing, visualization, feature engineering, feature selection, and machine learning model training.

Key findings:

- Age, BMI, glucose level, and smoking status significantly influence stroke prediction.
- Data balancing using SMOTE improves model performance.
- The Random Forest classifier provides reliable prediction accuracy.